

H19 One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

not yet rated

level: hypothesis · pipeline checks 86% · updated —

running. Exploratory round 1 done (2026-10-04): the pre-registered collapse P1 FAILED, with no common curve: equal-time activity gains rise with x_{att} while Hawkes talk gains fall, and the regime-only rival wins out of sample. Within the talk channel, the H...

WHAT HAPPENED

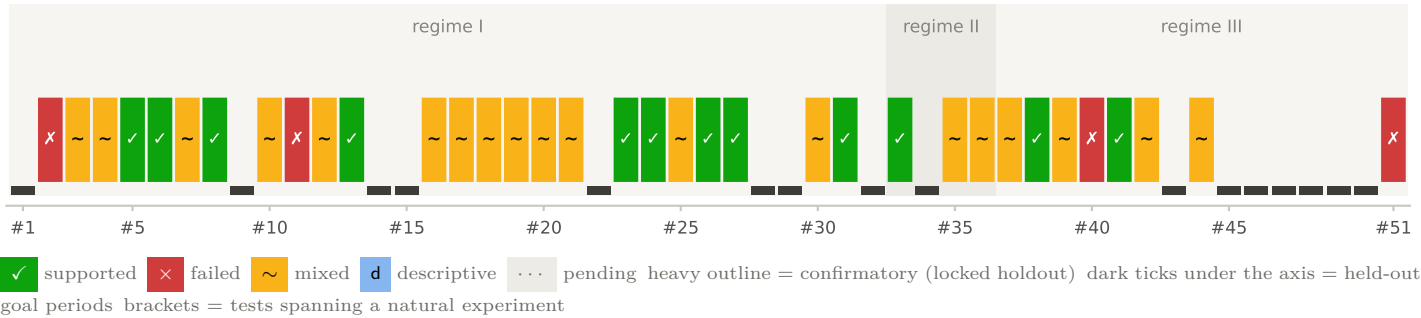
Not yet summarized. The question: Every hypothesis now produces per-goal-period coupling estimates: H02 Curie–Weiss βJ_0 , H03 branching ratio n and fast cross-triggering, H04 loop gain K , H05 J_{in}/J_{out} , and soon H10–H18. Do they collapse onto a single function of an operational control parameter, such as N , messages per agent-turn, hours per day, rooms, or the coupling mode of the goal? A collapse would be the swarm’s phase diagram and would predict coupling from operating conditions. Practical payoff: forecasting a swarm’s collective regime from its configuration.

PHYSICAL MODEL

From: physics-models/01-inverse-ising and 09-hawkes (mean-field loop gain $\beta J_0(1-m^2)$, branching ratio n), 02-nonequilibrium-ising. Finite-size scaling and data-collapse methodology. Hierarchical (partial-pooling) meta-regression across periods is the allowed exception (d) of the per-period rule: every period keeps its own estimate, and the meta-regression only relates those estimates to the period’s control parameters

Pending.

GOAL PERIODS TESTED 12 supported, 4 failed, 19 mixed



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.